



Using Simulations

Simulations are used to model real-world probability events and can be used to draw conclusions about population parameters.

Using Simulations to Draw Conclusions

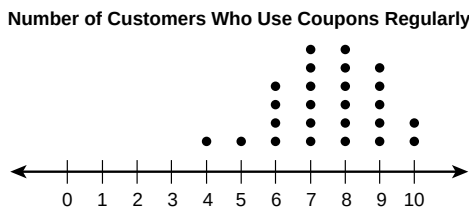
A **simulation** is a way to model random events. Simulations can be conducted to model the outcomes of real-world problems using methods such as flipping a fair coin, rolling number cubes, using a table of random digits, using a random number generator, and employing other methods.

The advantage of using a simulation is you can quickly gather data and use that information to draw conclusions about population parameters. You can also determine whether a statistical model is consistent with real-world results.

EXAMPLE 1

In a sample of grocery store customers who participated in a survey, 60% said that they use coupons on a regular basis. Based on nationwide market research, an analyst hypothesizes that the true population proportion is closer to 75%. To test whether this discrepancy is significant, he conducts a simulation assuming that the true population proportion is 75%. A random number generator is used to determine how many customers out of 10 would use coupons on a regular basis.

The line plot shows the results from 25 trials of the simulation in which the population proportion is 75%.



- A** Does the range of the line plot suggest that the random number generator is fair? Explain.

SOLUTION

The results of the simulation are clustered fairly tightly in the interval from 6 to 9 customers, with 7 and 8 customers being the most common result. The entire range of simulation results is from 4 to 10 customers. It would not be very likely for the simulation to result in 0 to 3 customers if the population proportion is 75%, so it is not unusual that this situation never occurred in the 25 trials. The random number generator appears to be fair.

- B** Does it seem likely that a proportion of 60% could have been achieved in the sample if the true population proportion is 75%? Explain.

SOLUTION

Out of 25 trials of the simulation, a result of 6 or fewer customers occurred 6 times. This situation represents an experimental probability of 24%, so it seems fairly likely that a random sample of customers could result in a proportion of 60% when the true population proportion is 75%.

- C** Do the results of the simulation indicate that the true population could be 75% despite the survey sample results? Explain.

SOLUTION

Yes, it could be possible for the population proportion to be 75%. The simulation results show that if the population proportion were 75%, you could get samples ranging from 40% to 100%. However, because most of the samples would fall in the interval from 60% to 90%, it is very possible that the true population proportion could be 75% despite the survey results. ■

► **Think About It** The simulation or experiment only accounts for a small sample size of possible outcomes. When using simulations to draw conclusions about population parameters, it is beneficial to conduct a large number of simulation trials to obtain a more accurate understanding of the possible samples.

Using Interval Estimates for Population Proportions

Interval estimates for a population proportion can be determined from data that you have gathered. If you collect a range of data values through a simulation or from a sample of a population, you can use this range to determine the interval estimate for the population proportion.

To determine an interval estimate for a population proportion, use the mean of the sample as the point estimate. Then use half the range of the sample as the margin of error.

Using Half the Range

interval estimate for p : mean $\pm$ half the range

EXAMPLE 2

A wildlife conservation officer estimates that about 65% of the ash trees in a region will be lost to the emerald ash borer infestation. She uses a table of random digits to simulate what percent of the trees will be lost. Suppose the mean of the sample proportions is 0.62, the least sample proportion is 0.45, and the greatest sample proportion is 0.77. Use half the range to determine an interval estimate for p , the percent of trees that will be lost to the infestation.

SOLUTION

Determine the range of the simulated sample proportions.

$$\text{range: } 0.77 - 0.45 = 0.32$$

Divide the range by 2 to determine half the range.

$$\text{half the range: } \frac{0.32}{2} = 0.16$$

Use the mean of the sample proportions and half the range as the margin of error to determine the interval estimate.

$$\text{interval estimate: } 0.62 \pm 0.16 = (0.46, 0.78)$$

Using half the range to determine the interval estimate, the simulation predicts that the percent of trees lost will be between 46% and 78%. ■

You can also use the standard deviation to determine an interval estimate for a population mean. Use the mean of the sample as the point estimate and twice the standard deviation as the margin of error.

Using the Standard Deviation

interval estimate for μ : mean $\pm 2 \times$ standard deviation

EXAMPLE 3

In a survey of 20 randomly selected members of a cycling club, members were asked how many miles they rode on the first day of a club riding event. The results were 22, 21, 28, 24, 23, 24, 20, 19, 25, 25, 22, 28, 28, 22, 27, 26, 28, 25, 24, and 25 mi. Use the standard deviation to determine an interval estimate for μ , the mean number of miles ridden by all of the club members.

SOLUTION

Use a graphing calculator or other technology to determine the mean of the sample.

mean: 24.3

Use a graphing calculator or other technology to determine the standard deviation of the sample. Round to two decimal places.

standard deviation: 2.74

Find the interval estimate for μ .

interval estimate: $24.3 \pm 2 \times 2.74 = (18.82, 29.78)$

Using the standard deviation to determine the interval estimate, the sample predicts that the mean number of miles ridden by members of the cycling club is between 18.82 and 29.78. ■